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Full Length Article

MSIFN: Multi-source information fusion network with dual-space constraints for few-shot classification

Hantao Wang [a], Huan Wan [b], Ghulan Mohiudin [a], Jianfeng Xu [a], Hong Rao [a], Xin Wei [a,*]

[a] School of Software, Nanchang University, Nanchang, Jiangxi, 330047, China
[b] School of Artificial Intelligence, Jiangxi Normal University, Nanchang, Jiangxi, 330022, China

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ABSTRACT

Few-shot learning poses a critical challenge due to insufficient feature discriminability and generalization caused by the limited availability of labeled samples. While previous methods have attempted to mitigate this by incorporating auxiliary information, they typically rely on a singular source, failing to capture a holistic view of visual concepts. This work investigates the strategic utilization and fusion of multi-source information. Empirical analysis demonstrates that such strategic utilization and fusion can effectively enhance few-shot learning performance, providing a promising direction for future exploration. We propose a novel Multi-Source Information Fusion Network (MSIFN). Unlike previous approaches, MSIFN comprehensively integrates heterogeneous information to construct a more expressive representation. In this network, we design a Semantic-Guided Selection Mechanism to facilitate the precise extraction of class-relevant information via a cross-modal autoencoder, ensuring the mitigation of negative transfer inherent in indiscriminate fusion. Furthermore, Dual-Space Constraints are imposed to counteract potential feature drift during fusion. By enforcing alignment with class prototypes in both visual and semantic spaces, this strategy ensures high intra-class compactness and semantic consistency. Extensive experiments on miniImageNet, tieredImageNet, and CUB benchmarks demonstrate that our method achieves superior performance compared to state-of-the-art approaches. The code for this work is available at: https://github.com/wanghantao-ncu/MSIFN.

1. Introduction

Deep learning has demonstrated remarkable efficacy in image classification, heavily relying on the availability of large-scale annotated datasets. However, in practical applications, acquiring such exhaustive supervision is often prohibitively expensive and impractical. In such cases, deep learning models typically suffer from severe overfitting and fail to generalize to novel categories. To alleviate this data dependency, Few-Shot Learning (FSL) has attracted strong research interest, aiming to empower models to rapidly adapt to new tasks with very few labeled examples.

In practice, due to the limited availability of labeled examples, the extracted features often fail to capture the intrinsic intra-class variability of a category. Consequently, the extracted feature representations are often insufficiently expressive and inherently biased toward the limited observed examples. To address these problems, incorporating auxiliary information is a straightforward technique. Existing methods typically focus on utilizing singular source of auxiliary information. A prominent stream of research leverages base-class priors to calibrate feature distributions. For instance, DC (Yang et al., 2021) calibrates the distribution of novel classes using base-class statistics, enabling the sampling of sufficient features for data augmentation. DDWM (Wei et al., 2023) proposes a direction-driven weighting strategy to align the biased distribution of novel classes with the real distribution by measuring the influence of different base classes. Similarly, ABC (Xu et al., 2025) posits that novel class centers reside in the voids between approximate base classes, thereby compensating for scarce samples by calibrating them toward these blank regions. Additionally, other approaches focuses on exploiting local discriminative details to augment global representations. FeatWalk (Chen et al., 2024) complements global features with local details, addressing the limitation that pre-trained extractors often fail to capture sufficient fine-grained information. TST (Sun et al., 2025) adopts a two-stage metric fusion learning framework, which promotes mutual enhancement between global and local features to obtain more robust representations. However, reliance on a singular information source hinders the holistic characterization of visual concepts. Consequently, the

* Corresponding author.
E-mail addresses: hantaowang@email.ncu.edu.cn (H. Wang), huanwan@jxnu.edu.cn (H. Wan), mohiuddin@ncu.edu.cn (G. Mohiudin), jianfeng_x@ncu.edu.cn (J. Xu), raohong@ncu.edu.cn (H. Rao), xinwei@ncu.edu.cn (X. Wei).

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strategic integration of heterogeneous knowledge is indispensable to counteract the bias inherent in data scarcity.

Although incorporating multi-source information theoretically compensates for the deficiencies of single-view representations, achieving its strategic utilization and fusion poses a challenge due to data heterogeneity and quality disparity. In contrast to well-annotated training data, auxiliary data often exhibit substantial noise, extending from irrelevant background clutter to semantic misalignment. Consequently, the indiscriminate assimilation or direct concatenation of such heterogeneous inputs inevitably induces negative transfer, thereby compromising the integrity and discriminability of the learned features. Thus, the challenge resides not in the mere integration of diverse data, but in devising a robust mechanism to selectively distill and synergistically fuse these inputs, ensuring the augmentation of target concepts while suppressing confounding noise.

In response to this challenge, we propose a novel Multi-Source Information Fusion Network (MSIFN). Different from previous works, our framework synergistically synthesizes global context, local details, base-class priors, and semantic embeddings to yield an expressive representation. Crucially, we incorporate a Semantic-Guided Selection Mechanism designed specifically to differentiate our approach from indiscriminate fusion. By leveraging the stability of semantic concepts, this mechanism employs a cross-modal autoencoder to project unlabeled visual patches into the semantic space. Such a design facilitates rigorous interaction among these sources, enabling the model to adaptively filter for semantic alignment with the target class, thereby effectively suppressing background clutter.

Moreover, the integration of heterogeneous information, while augmenting expressiveness, inevitably disrupts the intrinsic topology of the feature space. This distortion precipitates feature drift, causing the learned representations to deviate from their class prototypes. Prior works have investigated the imposition of constraints to enhance the discriminability of fused features. For instance, NormFace (Wang et al., 2017) introduces a class-center triplet loss to minimize the intra-class distance of deep visual features in face recognition. SGMA (Zhu et al., 2019) proposes a multi-attention loss that applies geometric constraints to encourage compact and diverse attention distributions. Similarly, SIFT (Pan et al., 2024) leverages calibrated prototypes, derived by integrating query cluster centers with support set prototypes, as anchors to enforce high intra-class compactness. However, such methods remain predominantly confined to the visual domain. To rectify the distributional deviation induced by multi-source fusion, we impose Dual-Space Constraints. By strictly aligning the fused features with stable prototypes in both visual and semantic spaces, our strategy ensures that the enhanced representations remain structurally compact and robust against noise.

Our main contributions include the following three aspects:.

- We explore the rational utilization and strategic fusion of multi-source information in few-shot learning. To this end, we propose the Multi-Source Information Fusion Network (MSIFN), a unified framework that synergistically integrates heterogeneous knowledge. Departing from indiscriminate fusion strategies, we devise a Semantic-Guided Selection Mechanism that facilitates fine-grained interaction to adaptively distill discriminative cues and suppress background clutter, thereby effectively mitigating negative transfer.
- We devise Dual-Space Constraints to rectify the feature drift induced by multi-source fusion. By aligning the fused representations with stable prototypes in both visual and semantic spaces, this strategy ensures high intra-class compactness and fortifies the feature distribution against deviations.
- We conduct extensive experiments on miniImageNet, tieredImageNet, and CUB benchmarks. The results not only demonstrate the superior performance of our method compared to state-of-the-art techniques but also empirically verify that the rational utilization and strategic fusion of multi-source information constitute an effective pathway for enhancing few-shot learning performance.

2. Related work

In few-shot learning, the extreme scarcity of training samples inevitably leads to biased feature representations. Driven by the need to mitigate this data deficiency, researchers have proposed incorporating various forms of auxiliary information. Existing methods can be broadly classified into three categories based on the source of auxiliary information: local discriminative features, base-class priors, and semantic knowledge.

Local Discriminative Features: Global pooling operations tend to discard fine-grained details essential for distinguishing visually similar classes. Realizing this limitation, many studies focus on leveraging local discriminative information to enhance feature representation. Li et al. (2019) introduce a Deep Nearest Neighbor Neural Network using a local-descriptor-based image-to-class measure to replace the image-level feature-based measure in the final layer. Ren et al. (2022) employ a multi-local feature relationship network to generate additional discriminative features via local feature attention between support and query sets. Wang et al. (2023c) propose QSFormer, which comprises a SampleFormer branch and a PatchFormer branch. Specifically, SampleFormer is designed to capture the inter-sample dependencies between the support and query sets for discriminative image representation, while PatchFormer extracts structural representations for each image sample by modeling the long-range dependencies of local image patches. Zhou and Cai (2024) propose a framework called global and local representation collaborative learning, which combines the complementary advantages of global equivariance and local aggregation. Zhang et al. (2024c) present a few-shot partial feature extraction network designed to flexibly learn critical partial descriptors. This approach captures distinguishable partial representations, thereby reconstructing the category-specific feature space and eliminating the negative implications of redundant features. Xiao et al. (2025) introduce a Fusion of Global and Adaptive Local Information (FGAL) method to address challenges in model optimization and intra-class variations by fully considering the beneficial impacts and interactions between global and local views. Although these approaches recover fine-grained cues that are often suppressed by global pooling, unfiltered local patches may also introduce task-irrelevant background content. Effective local-feature augmentation therefore requires selecting patches that are discriminative for the target class.

Base-Class Priors: Another prevalent strategy involves utilizing the abundant data or statistics from base classes to calibrate distributions or augment the support set for novel classes. Chen et al. (2019) demonstrate that when the domain discrepancy between base and novel classes is limited, training a feature extractor with a deep neural network backbone and employing a well-designed classifier yields highly competitive performance. Zhang et al. (2021) propose constructing a pseudo-feature library for training sample augmentation. Specifically, they perform multiple samplings on base-class features to establish a distribution-level pseudo-feature library, while supplementing it with a patch-level pseudo-feature library via image cropping operations. Wang et al. (2023b) find that the feature extractor trained on base classes can represent the semantic similarity between base and unseen novel classes. They propose a simple yet effective Training-frEE Calibration (TEEN) strategy to enhance the discriminability of new classes by fusing new prototypes with weighted base prototypes. Xu et al. (2025) introduce an Adaptive Blank Compensation method that learns more accurate class distributions to improve the accuracy of few-shot classification based on data enhancement. Distribution Calibration (DC) (Yang et al., 2021) and DDWM (Wei et al., 2023) leverage base-class statistics to calibrate novel-class feature distributions. Taken together, these methods transfer base-class knowledge through prototype calibration, feature augmentation, or distribution calibration. The utility of the transferred knowl-

edge depends on the semantic and visual relatedness between base and novel classes. Mismatched base priors may distort novel-class representations and induce negative transfer. MSIFN therefore employs a coarse-to-fine retrieval strategy that first identifies semantically related base classes and then retrieves visually similar instance-level features from the matched class.

Semantic Knowledge Embeddings: Realizing that visual information alone is insufficient under extreme data scarcity, researchers have integrated semantic priors to guide the learning process. Schwartz et al. (2022) focus on the role of additional semantics in significantly facilitating few-shot visual learning, advocating for the combination of multiple and richer semantics (category labels, attributes, and natural language descriptions). Yang et al. (2022) propose a Semantic Guided Attention mechanism that maps semantic knowledge to category-specific attention vectors, which are then used to perform feature selection and enhance visual prototypes. Cao and Zhang (2022) present a novel semantic alignment model for relation comparison, which introduces a semantic alignment loss to align intra-class feature relation statistics and incorporates local-global mutual information maximization to learn representations with locally consistent and intra-class shared information across image structural locations. Yang et al. (2023) introduce a semantics-guided latent parts embedding method, which generates class-specific latent part filters using semantic knowledge to extract local features. Zhang et al. (2024b) design a simple and efficient way to translate high-quality semantics and visual features into prototypes, without any intricate semantic understanding modules. Pan and Shen (2024) propose the CMDD framework to mitigate deviations in both feature representations and prototypes within few-shot learning scenarios. Specifically, the framework leverages class semantic information to assign labels to query cluster centers, thereby facilitating more accurate and consistent prototype refinement. Li et al. (2025c) introduce a Text-Guided-Image (TGI) component to mitigate the anomalous match of image modality through text representations. These methods draw on heterogeneous semantic sources, including static word embeddings and pretrained language or vision-language models. Baby Steps (Schwartz et al., 2022) combines GloVe embeddings of class names with textual descriptions encoded using BERT (Devlin et al., 2019). SemFew (Zhang et al., 2024b) augments WordNet definitions with GPT-3.5 and encodes the resulting descriptions using CLIP (Radford et al., 2021) or BERT (Devlin et al., 2019). Because these approaches differ in both external knowledge sources and evaluation protocols, we compare their mechanisms with MSIFN rather than rank their reported accuracies. Most existing methods use semantic information primarily for prototype construction, attention, or cross-modal alignment. By contrast, MSIFN uses dataset-specific semantic priors both to assess the relevance of base classes and local patches and to regularize the fused representation.

Taken together, prior work shows that local cues, base-class priors, and semantic knowledge provide complementary information for few-shot classification. However, these sources are often exploited in isolation or fused without explicit relevance estimation, which may propagate auxiliary noise and induce representation drift. To address these issues, we propose the Multi-Source Information Fusion Network (MSIFN). MSIFN uses semantic guidance to select task-relevant local and base-class features, integrates them with the global representation, and regularizes the resulting representation in both visual and semantic spaces.

3. Methodology

3.1. Problem definition

In the setting of few-shot image classification, let $\mathcal{D} = \{(x_i, y_i)\}$ denote the entire dataset, where $x_i \in \mathbb{R}^D$ represents the D-dimensional feature vector of the ith image, and $y_i \in \mathcal{Y}$ is its corresponding class label. The dataset $\mathcal{D}$ is divided into two disjoint subsets: a base dataset $\mathcal{D}_{base}$ containing base classes $\mathcal{Y}_{base}$, and a novel dataset $\mathcal{D}_{novel}$ containing novel classes $\mathcal{Y}_{novel}$, such that $\mathcal{Y}_{base} \cap \mathcal{Y}_{novel} = \emptyset$. The primary objective is to train a model on $\mathcal{D}_{base}$ that generalizes well to unseen classes in $\mathcal{D}_{novel}$. We adhere to the standard N-way K-shot episodic paradigm. In each episode, a task is constructed by randomly sampling N classes from the available class set. For each sampled class, K labeled instances are selected to constitute the support set $\mathcal{S}$, while M distinct instances serve as the query set $\mathcal{Q}$ for evaluation. Unlike traditional few-shot settings, we explicitly incorporate semantic priors into the support set. Consequently, the support set is formulated as $\mathcal{S} = \{(x_i, y_i, s_{y_i})\}_{i=1}^{N\times K}$, where x_i denotes the visual sample, y_i is the class label, and s_{y_i} represents the semantic embedding of class y_i, extracted via the GloVe (Pennington et al., 2014) model. The query set is defined as $\mathcal{Q} = \{(x_j, y_j)\}_{j=1}^{N\times M}$, where the ground-truth labels y_j are withheld during inference. The goal of our method is to construct a robust support set by strategically fusing heterogeneous information. Subsequently, a classifier is trained on these fused representations to predict the labels of query samples. Finally, the classification performance is evaluated over extensive testing tasks.

3.2. Overview

As illustrated in Fig. 1, the proposed Multi-Source Information Fusion Network (MSIFN) is devised to mitigate the intrinsic bias of few-shot learning through the rational utilization and strategic fusion of heterogeneous information. To avoid the risks of indiscriminate fusion, the framework is structured into three modules, progressing from semantic bridging to adaptive selection, and finally to robust fusion: (1) Cross-Modal Autoencoder Construction. A unified representational space is the prerequisite for effective selection. Addressing the lack of explicit semantic labels in local patches, we first establish a Cross-Modal Autoencoder to bridge the visual and semantic domains. By learning a visual-to-semantic projection, this module enables the model to decode the "latent semantics" of diverse local patches, establishing a robust basis for relevance assessment. (2) Semantic-Guided Information Selection. To mitigate negative transfer induced by noise, we devise a Semantic-Guided Selection Mechanism. Exploiting the pre-trained autoencoder, this module quantifies the semantic consistency between auxiliary information and the target class. This process adaptively distills discriminative cues while suppressing confounding noise, ensuring that only high-quality information is retained for integration. (3) Multi-Source Fusion and Dual-Space Constraints. The final stage involves synthesizing a holistic representation via a Transformer-based architecture. Simultaneously, to rectify the feature drift inherent in adaptation, we enforce Dual-Space Constraints. This strategy aligns the fused features with stable prototypes in both visual and semantic spaces, thereby preserving distributional stability and enhancing intra-class compactness.

3.3. Cross-modal autoencoder training

To establish a bidirectional mapping between the visual and semantic domains, we construct a cross-modal autoencoder comprising two components: an encoder W_e that projects visual features into the semantic space, and a decoder W_d that reconstructs visual features from semantic embeddings. The model is optimized via a joint objective function that simultaneously minimizes the visual reconstruction loss and the semantic alignment loss:

$$\min \mathcal{L}_{aut} = \sum_{i=1}^{N\times K} \|x_i - \tilde{x}_i\|^2 + \lambda \|s_i - \tilde{s}_i\|^2, \quad \text{with} \quad \tilde{s}_i = W_e x_i, \quad \tilde{x}_i = W_d s_i. \tag{1}$$

Crucially, to prevent detrimental domain shifts that would arise from utilizing base-class data, this autoencoder is optimized exclusively on the support set S during each episodic task. Here, $\tilde{s}_i$ is the predicted semantic embedding generated by the encoder, while $\tilde{x}_i$ is the visual embedding reconstructed from the ground-truth semantic feature s_i via the decoder. The hyperparameter λ balances the trade-off between the

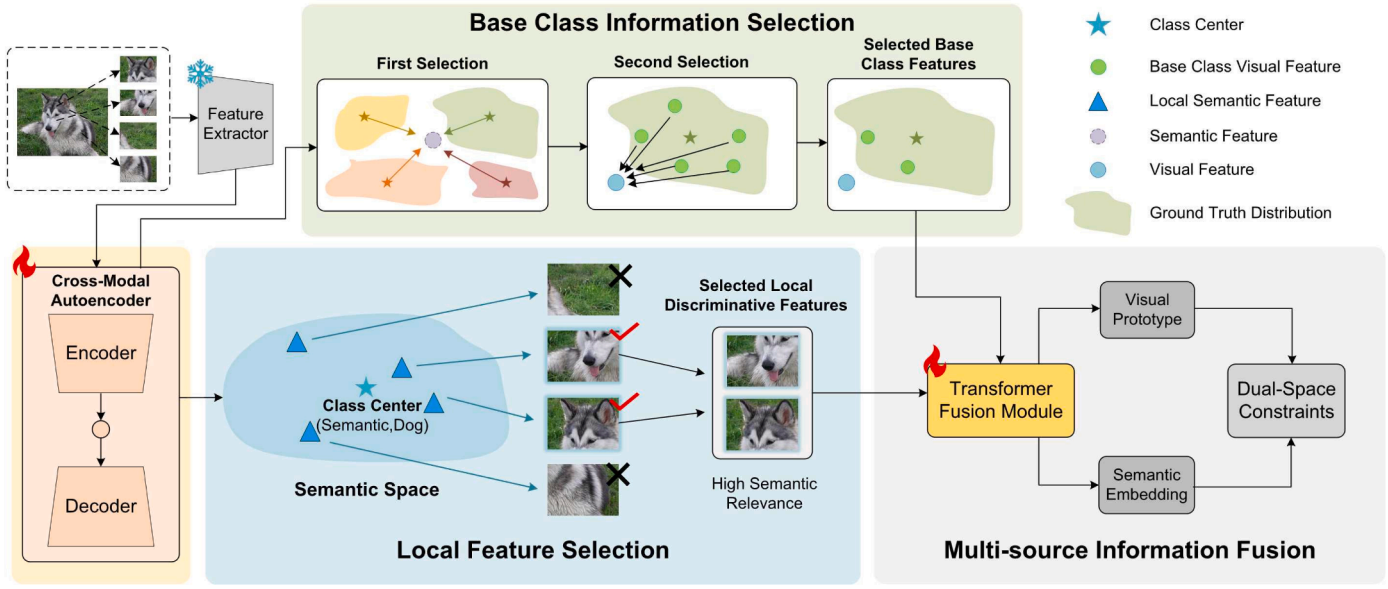


Fig. 1. An overview of the MSIFN framework. The architecture comprises three key components: (1) a cross-modal autoencoder trained to extract semantic embeddings; (2) a Semantic-Guided Selection Mechanism employed to filter class-relevant information; and (3) a Multi-Source Fusion module designed to integrate selected multi-source cues and apply Dual-Space Constraints to regularize the fused features.

visual reconstruction task and the semantic alignment task. In the experiments, we set $\lambda = 0.5$ for all the datasets.

3.4. Multi-source information selection

The indiscriminate integration of heterogeneous multi-source information risks inducing negative transfer due to disparate data quality and inherent noise. Moreover, feature selection based exclusively on visual support sets is intrinsically fragile; given the paucity of labeled samples, such visual prototypes are susceptible to severe distributional bias and often deviate from the true category distribution. To overcome this challenge, we exploit robust class semantic information as a stable, unbiased prior, and propose a Semantic-Guided Information Selection Mechanism.

3.4.1. Base class information selection

The abundant annotated data in the base dataset serves as a substantial reservoir of prior knowledge. However, given the disjoint label space, the indiscriminate transfer of these priors inherently risks inducing semantic ambiguity and detrimental domain shifts. To resolve this conflict and ensure rational utilization, we devise a hierarchical selection strategy that operates in a coarse-to-fine manner, progressing from global semantic alignment to fine-grained visual retrieval.

Class-Level Semantic Alignment. Given the scarcity of support samples, visual prototypes are inherently susceptible to bias, making them unreliable for global auxiliary information retrieval. Consequently, we leverage the semantic space as a stable anchor to establish a robust alignment between novel and base classes. We formulate this alignment task as a global bipartite matching problem. Crucially, unconstrained nearest-neighbor retrieval often causes multiple novel classes to collapse toward a single dominant base class. By imposing a strict one-to-one exclusivity constraint, our formulation explicitly prevents this semantic ambiguity, strictly preserving the topological distance margins among novel classes and effectively eliminating a primary source of negative transfer. The optimization objective maximizes the total semantic similarity under this exclusivity constraint, modeled as the following binary linear programming problem:

$$\begin{aligned} \max \quad & \sum_{i=1}^{N} \sum_{j=1}^{|C_{base}|} p_{ij} z_{ij} \\ \text{s.t.} \quad & \sum_{j=1}^{|C_{base}|} z_{ij} = 1, \quad \forall i \in \{1, \dots, N\}, \\ & \sum_{i=1}^{N} z_{ij} \le 1, \quad \forall j \in \{1, \dots, |C_{base}|\}, \\ & z_{ij} \in \{0, 1\}. \end{aligned} \tag{2}$$

Here, p_{ij} denotes the cosine similarity between the semantic embedding of the ith novel class s_i^n and the jth base class s_j^b, calculated as $p_{ij} = \frac{s_i^n \cdot s_j^b}{\|s_i^n\| \cdot \|s_j^b\|}$. The binary decision variable $z_{ij} \in \{0, 1\}$ indicates whether base class j is assigned to novel class i. Solving this optimization yields the matched base class index $\phi(i) = \{j \mid z_{ij} = 1\}$ for each novel class.

Mathematically, the semantic alignment in Eq. (2) is formulated as a constrained combinatorial optimization problem. In standard few-shot retrieval, unconstrained nearest-neighbor search often suffers from the "hubness" problem, where multiple novel classes collapse toward a single dominant base class. This phenomenon effectively shrinks the decision margins between novel categories, leading to severe negative transfer. By explicitly imposing the one-to-one exclusivity constraint through bipartite matching ($\sum z_{ij} \le 1$), our mechanism guarantees that the augmented feature space preserves the original topological separation of novel classes. This optimization-based selection ensures that the retrieved knowledge is not only semantically relevant but also maintains the structural integrity of the classification boundaries.

Instance-Level Visual Selection. While the selected base class guarantees semantic consistency, substantial intra-class visual variation persists, potentially inducing detrimental distribution shifts. Consequently, reliance on coarse class-level labels is inadequate. To bridge this gap, we implement a fine-grained retrieval strategy within the visual space. Specifically, given a support sample x, we retrieve a subset of nearest neighbors $\mathcal{X}^b$ from the candidate base set $\mathcal{D}_{base}^{\phi(i)}$. This is accomplished by computing the squared Euclidean distance between global features and selecting the top-B instances exhibiting the smallest distances:

$$\begin{aligned} \mathcal{X}^b &= \left\{ x_m^b \in \mathcal{D}_{base}^{\phi(i)} \mid m \in \mathcal{I}_{\text{base}} \right\}, \\ \mathcal{I}_{\text{base}} &= \arg\text{topB}\left(-\|x - x_m^b\|_2^2, B\right). \end{aligned} \tag{3}$$

Here, $\mathcal{I}_{\text{base}}$ denotes the index set of the selected base instances. The arg topB operator identifies the indices of the B samples with the largest

values. Mathematically, this retrieves the nearest neighbors by casting distance minimization as a maximization problem via the negation of the squared L_2 norm. By conditioning the retrieval on the specific visual characteristics of x, this strategy ensures that the supplementary set $\mathcal{X}^b$ preserves high visual coherence, thereby effectively mitigating negative transfer.

3.4.2. Local feature selection

Global pooling operations inherently aggregate spatial dimensions, inevitably discarding fine-grained cues critical for distinguishing visually similar few-shot categories. While incorporating local patches compensates for this information loss, it presents a significant trade-off: the inclusion of local context often introduces task-irrelevant background noise. The unconstrained fusion of such distractors risks inducing negative transfer. To address this challenge, we devise a semantic-guided mechanism that adaptively disentangles object-relevant patterns from background interference.

Latent Semantic Mapping. The absence of explicit semantic supervision for local patches impedes effective filtering. To bridge this modality gap, we leverage the pre-trained Cross-Modal Autoencoder as a visual-semantic interface. Let $\mathcal{P}$ denote the set of all candidate local patches extracted from the support image. By projecting each patch $x_l \in \mathcal{P}$ into the shared latent semantic space via $\tilde{s}_l = W_e x_l$, we endow the raw visual segments with interpretable semantic embeddings.

Discriminative Patch Filtering. To isolate discriminative regions, we quantify the relevance of each patch by evaluating its alignment with the target class semantics. Specifically, we compute the cosine similarity between the projected latent vector $\tilde{s}_l$ and the ground-truth class embedding s_i. Serving as a semantic filter, this metric retains only the subset of patches $\mathcal{X}^l$ that demonstrate strong semantic affinity with the target prototype:

$$\begin{aligned} &\mathcal{X}^l = \{x_l \in \mathcal{P} \mid l \in \mathcal{I}_{\text{local}}\}, \\ &\mathcal{I}_{\text{local}} = \arg\text{topL}\left(\frac{\tilde{s}_l^\top s_i}{\|\tilde{s}_l\|_2 \|s_i\|_2}, L\right), \\ &\text{s.t.} \quad \tilde{s}_l = W_e x_l, \quad \forall x_l \in \mathcal{P}. \end{aligned} \tag{4}$$

Here, $\mathcal{I}_{\text{local}}$ represents the index set returned by the $\arg\text{topL}$ operator, which identifies the indices of the L patches with the highest similarity scores. Consequently, the final local feature set $\mathcal{X}^l$ comprises fine-grained patterns that are strictly aligned with the target category, effectively suppressing background noise.

The local feature selection process can be formally interpreted through the principle of mutual information maximization. Unconstrained integration of local patches substantially inflates the dimensionality and noise variance of the input sequence, an issue particularly detrimental in data-scarce regimes. By projecting these patches into the latent semantic space and applying a hard thresholding operator based on cosine similarity (Eq. (4)), the proposed mechanism implicitly maximizes the mutual information between the retained local descriptors X^l and the ground-truth class label y_i. This deterministic filtering strictly limits integration to regions possessing both high semantic affinity and concentrated task-relevant information density. Consequently, this operation effectively constrains the noise variance prior to the Transformer-based fusion stage.

3.5. Multi-source information fusion

3.5.1. Transformer-based multi-source fusion

Following the discriminative filtering of auxiliary information, our aim is to synthesize these heterogeneous sources into a holistic representation. To effectively model the complex dependencies and intrinsic correlations among the global context, local discriminative details, and base-class priors, we employ a Transformer-based fusion module. Adopting the tokenization paradigm of Vision Transformers (ViT) (Dosovitskiy, 2020), we treat the calibrated multi-source information as a unified sequence of tokens. Formally, for a support sample x_i, we construct the input sequence $\mathcal{T}_{in}$ by concatenating its global feature x_i with the retrieved base-class features $\mathcal{X}^b$ and the refined local patches $\mathcal{X}^l$:

$$\mathcal{T}_{in} = [x_i, \mathcal{X}^b, \mathcal{X}^l] \in \mathbb{R}^{(1+B+L)\times D}, \quad \text{where } x_i \in \mathcal{S}. \tag{5}$$

This composite sequence is fed into a single-layer Transformer encoder. Within this module, the multi-head self-attention mechanism dynamically captures long-range dependencies across these heterogeneous sources, adaptively modulating the contribution of each component based on its semantic relevance. Finally, we extract the transformed token corresponding to the global feature x_i as the final fused representation $\tilde{x}_i$:

$$\tilde{x}_i = \text{Transformer}(\mathcal{T}_{in}). \tag{6}$$

This process yields a robust feature representation that integrates holistic and fine-grained cues while suppressing noise via the attention mechanism.

3.5.2. Dual-space constraints

While multi-source fusion enhances feature expressiveness, it inevitably perturbs the original feature distribution, introducing a risk of feature drift where learned representations may deviate from the class prototype. To counteract this instability and enforce high intra-class compactness, we impose Dual-Space Constraints. As illustrated in Fig. 2, this strategy explicitly aligns the fused features with stable prototypes in both visual and semantic spaces.

Visual Space Constraint. We leverage the original global features to establish stable class prototypes. This mechanism functions as a robust regularizer for the fusion module, ensuring that the synthesized representations preserve the geometric structure of the visual space. Specifically, for each class k, the class prototype p_k is computed as the centroid of the support features belonging to that class:

$$p_k = \frac{1}{|\mathcal{S}_k|} \sum_{x \in \mathcal{S}_k} x, \tag{7}$$

where $\mathcal{S}_k$ denotes the support set for class k. We formulate the visual constraint loss $\mathcal{L}_v$ to minimize the squared Euclidean distance between each fused representation and its corresponding ground-truth class prototype:

$$\mathcal{L}_v = \sum_{i=1}^{N\times K} \|\tilde{x}_i - p_{y_i}\|_2^2, \tag{8}$$

where $\tilde{x}_i$ denotes the fused feature, and p_{y_i} represents the prototype corresponding to the ground-truth label y_i of sample x_i. By anchoring the fused features to these original representations, we enable the network to enhance discriminability while preventing significant deviation from the robust initial class distribution.

Semantic Space Constraint. In parallel, we impose constraints within the semantic space to ensure feature consistency across modal-

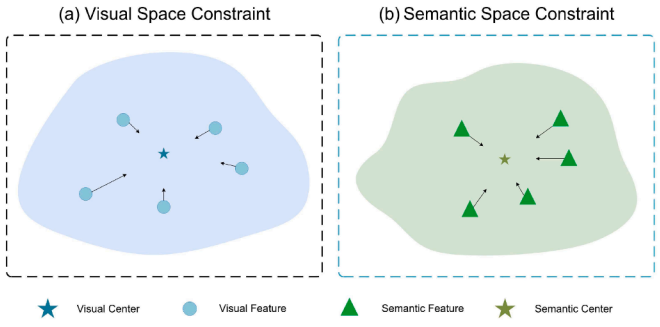


Fig. 2. Dual-space constraints. (a) Illustration of the visual feature constraint in the visual space for a 5-shot task. (b) Illustration of the semantic feature constraint in the semantic space for a 5-shot task.

ities. Leveraging the pre-trained encoder W_e, we project the fused feature $\tilde{x}_i$ into the latent semantic space via $\tilde{s}_i = W_e\tilde{x}_i$. We then align this predicted representation with the corresponding ground-truth semantic embedding s_{y_i}:

$$\mathcal{L}_s = \sum_{i=1}^{N\times K} \|\tilde{s}_i - s_{y_i}\|_2^2, \tag{9}$$

where s_{y_i} denotes the ground-truth semantic embedding for the class of instance x_i. Under the dual guidance of visually calibrated prototypes and ground-truth semantic embeddings, the network is driven to learn representations that converge toward stable class centroids. This explicit alignment effectively mitigates the distributional bias inherent in the original support set.

Total Objective. The joint enforcement of the Visual Proto Constraint and Semantic Proto Constraint addresses the fundamental bias-variance tradeoff inherent in few-shot adaptation. While integrating heterogeneous multi-source information enhances feature expressiveness, it inevitably perturbs the original feature distribution, precipitating feature drift. To counteract this instability, we introduce two complementary regularization terms. Specifically, the Visual Proto Constraint ($\mathcal{L}_v$) functions as a geometric regularizer designed to preserve high fidelity to the task-specific visual manifold. By anchoring the fused features to the empirical mean of the support set, this constraint strictly minimizes intra-class variance based on the observed visual data. However, because empirical visual prototypes act as high-variance estimators under extreme data scarcity, we simultaneously enforce the Semantic Proto Constraint ($\mathcal{L}_s$). This term introduces a zero-variance conceptual anchor to the optimization process. Unlike empirical visual features, these pre-trained embeddings are conceptually invariant and entirely immune to data scarcity.

The final objective function is formulated as a weighted sum of these visual and semantic regularization terms:

$$\mathcal{L} = w_1\mathcal{L}_v + w_2\mathcal{L}_s, \tag{10}$$

where w_1 and w_2 are hyperparameters balancing the two constraints. Theoretically, optimizing Eq. (10) strictly regularizes the latent representation space of the fused features. By bounding the hypothesis space with both a geometric anchor (empirical visual priors) and a conceptual anchor (invariant semantic embeddings), this synergistic mechanism forces the learned representations to converge precisely at the intersection of the task-specific visual manifold and the stable semantic space. This joint objective fundamentally suppresses feature drift, preventing deep representations from degenerating into high-variance, uninformative sub-spaces, thereby ensuring robust generalization to novel categories. Following this constrained fusion process, a standard Logistic Regression (LR) classifier is trained on the refined support features to predict the labels of query instances.

4. Experiments

4.1. Datasets

We evaluate the efficacy of the proposed method on three widely adopted few-shot classification benchmarks: miniImageNet (Ravi & Larochelle, 2017), tieredImageNet (Ren et al., 2018), and CUB-200-2011 (Wah et al., 2011).

miniImageNet serves as the standard benchmark in the field. Derived from ILSVRC-2012 (Russakovsky et al., 2015), it comprises 100 classes with 600 images per class, totaling 60,000 samples. Adhering to the standard division method, we partition the dataset into 64, 16, and 20 classes for training, validation, and testing, respectively.

To assess performance on a larger scale with greater semantic diversity, we utilize tieredImageNet. Also a subset of ILSVRC-2012, this dataset features a hierarchical structure that groups its 608 classes into broader super-categories. Crucially, the data split ensures that training, validation, and testing sets originate from disjoint super-categories, thereby creating a larger semantic gap that rigorously tests the model's generalization capability. We follow the standard partition of 351, 97, and 160 classes for training, validation, and testing.

Additionally, to evaluate performance on fine-grained discrimination, we employ the CUB-200-2011 dataset. It contains 11,788 images spanning 200 distinct bird species. We divide the classes into 100 for training, 50 for validation, and 50 for testing. In terms of preprocessing, we adopt resolution settings tailored to the specific characteristics of each benchmark. For miniImageNet and tieredImageNet, input images are resized to a resolution of 84×84 pixels. Conversely, for the fine-grained CUB dataset, images are resized to 224×224 pixels to preserve intricate visual patterns essential for distinguishing subordinate categories.

4.2. Implementation details

To ensure a fair comparison with state-of-the-art methods, we employ two standard backbones: ResNet-12 (Tian et al., 2020; Xie et al., 2022) and ResNet-18 (Liu et al., 2020). Consistent with established few-shot learning conventions, we adopt architecture-specific input resolutions: 84×84 pixels for ResNet-12 and 224×224 pixels for ResNet-18. Regarding data preparation, we generate 16 random crops per image to capture local details, with each crop covering 20% of the original spatial dimensions. To ensure strict reproducibility during evaluation, a fixed global random seed is applied to both task sampling and crop generation. The top-L candidates are subsequently selected via nearest neighbor retrieval. miniImageNet and tieredImageNet use 300-dimensional GloVe embeddings obtained by averaging the class-label word vectors, while CUB uses 312-dimensional human-annotated attributes. Furthermore, we explicitly incorporate auxiliary base-class information by retrieving the top-B nearest neighbors from the base dataset.

During the pre-training phase, we adopt the DeepBDC framework (Xie et al., 2022) to optimize the ResNet-12 and ResNet-18 feature extractors on the base-class training set using standard cross-entropy loss. Specifically, the ResNet-12 backbone comprises four stages with channel dimensions of $[64, 160, 320, 640]$. Each residual block contains three 3×3 convolutional layers, each sequentially followed by batch normalization and a LeakyReLU activation with a negative slope of 0.1. Spatial downsampling is executed via 2×2 max pooling after the first three stages. Crucially, in the final stage of both backbones, we replace the standard global average pooling layer with a Brownian Distance Covariance (BDC) pooling layer to generate the final image representations.

The networks are optimized using Stochastic Gradient Descent (SGD) with a momentum of 0.9, a weight decay of 5×10^{-4}, and a mini-batch size of 64. Following the initial pre-training, a sequential self-distillation strategy is applied on the base training set to further refine the representations. The resulting distilled model subsequently serves as our definitive feature extractor. Notably, this backbone remains strictly frozen during the subsequent few-shot evaluation phase, and is deployed directly to extract features across the training, validation, and test splits.

The proposed MSIFN architecture comprises two key modules. First, the Cross-Modal Autoencoder consists of an encoder and a decoder, both designed as single fully connected layers. Because the inputs are high-level deep visual features and pre-trained semantic embeddings rather than raw pixels, this exceptionally lightweight architecture enables rapid, task-specific adaptation on the limited support set (only $N \times K$ samples) while fundamentally mitigating the risk of severe overfitting. This module is optimized using the Adam optimizer with an initial learning rate of 1×10^{-4}, which is decayed by a factor of 0.2 every 5 steps. Second, the Multi-Source Fusion module is composed of a single-layer Transformer encoder. It is similarly optimized via Adam, with learning rates and iteration schedules calibrated to each specific dataset. Following the task-wise adaptation protocol of SIFT (Pan et al., 2024), we report the mean classification accuracy over a fixed sequence of 2000 tasks sampled from the testing split, accompanied by descriptive

Table 1
5-way 1-shot and 5-way 5-shot classification accuracy (%) on miniImageNet and tieredImageNet with 95% confidence intervals. The numbers in bold have intersecting confidence intervals with the most accurate method.

Methods	Backbone	miniImageNet		tieredImageNet	
		5-way 1-shot	5-way 5-shot	5-way 1-shot	5-way 5-shot
Meta-AdaM (Sun & Gao, 2023)	ResNet-12	59.89 ± 0.49	77.92 ± 0.43	65.31 ± 0.48	85.24 ± 0.35
DSFN (Zhang & Huang, 2022)	ResNet-12	61.27 ± 0.71	80.13 ± 0.17	65.46 ± 0.70	82.41 ± 0.53
ProtoNet (Snell et al., 2017)	ResNet-12	62.59 ± 0.85	78.60 ± 0.16	68.37 ± 0.23	83.43 ± 0.16
MetaOptNet (Lee et al., 2019)	ResNet-12	62.64 ± 0.61	78.63 ± 0.46	65.99 ± 0.72	81.56 ± 0.53
MatchNet (Vinyals et al., 2016)	ResNet-12	63.08 ± 0.80	75.99 ± 0.60	68.50 ± 0.92	80.60 ± 0.71
MetaBaseline (Chen et al., 2021)	ResNet-12	63.17 ± 0.23	79.26 ± 0.17	68.62 ± 0.27	83.74 ± 0.18
PBML (Fu et al., 2024)	ResNet-12	63.60 ± 0.70	81.94 ± 0.44	70.64 ± 0.72	85.39 ± 0.40
LSFSL (Padmanabhan et al., 2023)	ResNet-12	64.67 ± 0.49	81.79 ± 0.18	71.17 ± 0.52	86.23 ± 0.22
MetaDiff (Zhang et al., 2024a)	ResNet-12	64.99 ± 0.77	81.21 ± 0.56	72.33 ± 0.92	86.31 ± 0.62
DeepEMD (Zhang et al., 2020)	ResNet-12	65.91 ± 0.82	82.41 ± 0.56	71.16 ± 0.87	86.03 ± 0.58
FRN (Wertheimer et al., 2021)	ResNet-12	66.45 ± 0.19	82.83 ± 0.13	71.16 ± 0.22	86.01 ± 0.15
MELR (Fei et al., 2021)	ResNet-12	67.40 ± 0.43	83.40 ± 0.28	72.14 ± 0.51	87.01 ± 0.35
GEL (Wang et al., 2023a)	ResNet-12	68.26 ± 0.85	83.05 ± 0.55	70.50 ± 0.93	84.60 ± 0.65
ELMOS (Lai et al., 2023)	ResNet-12	69.32 ± 0.46	84.65 ± 0.32	73.14 ± 0.50	86.88 ± 0.36
DeepBDC (Xie et al., 2022)	ResNet-12	67.83 ± 0.43	85.45 ± 0.29	73.82 ± 0.47	89.00 ± 0.40
BANs-SLA (Liu et al., 2024)	ResNet-12	70.40 ± 0.44	85.31 ± 0.22	73.85 ± 0.49	87.72 ± 0.33
CPL-SPM (Dong et al., 2025a)	ResNet-12	69.44 ± 0.46	84.80 ± 0.31	76.32 ± 0.49	88.29 ± 0.31
FeatWalk (Chen et al., 2024)	ResNet-12	70.21 ± 0.44	87.38 ± 0.27	75.25 ± 0.48	89.92 ± 0.29
PRSN (Dong et al., 2025b)	ResNet-12	71.09 ± 0.90	86.82 ± 0.61	74.14 ± 0.86	88.54 ± 0.61
MSIFN (Ours)	ResNet-12	**87.08 ± 0.58**	**91.36 ± 0.39**	**78.50 ± 0.90**	**90.04 ± 0.56**

Table 2
Comparison with state-of-the-art few-shot classification methods on the CUB dataset in both 5-way 5-shot and 5-way 1-shot settings. The numbers in bold have intersecting confidence intervals with the most accurate method.

Methods	Backbone	CUB	
		5-way 1-shot	5-way 5-shot
ProtoNet (Snell et al., 2017)	Conv4	64.42 ± 0.48	81.82 ± 0.35
MAML (Finn et al., 2017)	ResNet-18	68.42 ± 1.07	83.47 ± 0.62
Baseline++ (Chen et al., 2019)	ResNet-18	67.02 ± 0.90	83.58 ± 0.54
MELR (Fei et al., 2021)	Conv4	70.26 ± 0.50	85.01 ± 0.32
AA (Afrasiyabi et al., 2020)	ResNet-18	74.22 ± 1.09	88.65 ± 0.55
Neg-Cosine (Liu et al., 2020)	ResNet-18	72.66 ± 0.85	89.40 ± 0.43
AAP2S (Ma et al., 2022)	ResNet-18	77.64 ± 0.19	90.43 ± 0.18
RTDC (Li et al., 2025a)	ResNet-18	77.82 ± 0.22	89.38 ± 0.11
FRN (Wertheimer et al., 2021)	ResNet-18	82.55 ± 0.19	92.98 ± 0.10
DeepBDC (Xie et al., 2022)	ResNet-18	84.01 ± 0.42	94.02 ± 0.24
ITAM (Li et al., 2025b)	ResNet-18	84.09 ± 0.18	93.41 ± 0.10
FeatWalk (Chen et al., 2024)	ResNet-18	85.67 ± 0.38	95.44 ± 0.16
MSIFN (Ours)	ResNet-18	**86.91 ± 0.78**	**95.58 ± 0.28**

95% confidence intervals. The task-adaptive modules retain the parameters learned from preceding episodes, whereas the LR classifier is fitted independently for each task. Each task involves 15 randomly sampled query images per class and is evaluated under both 5-way 1-shot and 5-way 5-shot settings.

4.3. Comparison with state-of-the-art methods

We present a comprehensive comparative analysis of MSIFN against state-of-the-art methods. Table 1 summarizes the quantitative results on the generic object recognition benchmarks, miniImageNet and tieredImageNet. As observed, MSIFN establishes a new state-of-the-art across all evaluated settings. The performance gains are particularly pronounced on the miniImageNet benchmark. In the challenging 5-way 1-shot setting, MSIFN attains an accuracy of 87.08%, surpassing the runner-up method (PRSN) by a substantial margin of approximately 16%. Similarly, in the 5-shot setting, we achieve 91.36%, outperforming prior approaches by roughly 4%. On the larger tieredImageNet dataset, MSIFN consistently maintains its superiority, yielding accuracies of 78.50% and 90.04% for 1-shot and 5-shot tasks, respectively. The performance evaluation on the fine-grained CUB benchmark is detailed in Table 2. In the data-scarce 1-shot scenario, our method outperforms the leading FeatWalk (Chen et al., 2024) algorithm, establishing a new state-of-the-art accuracy of 86.91%. This result empirically validates the efficacy of our semantic fusion strategy in discerning subtle inter-class discrepancies. Crucially, this superiority extends to the 5-way 5-shot setting, where MSIFN achieves a peak accuracy of 95.58%, surpassing the previous best performance of 95.44%. These results demonstrate that our framework consistently maintains superior performance and robustness across diverse data regimes.

4.4. Ablation study

4.4.1. Analysis of multi-source

Table 3 presents a comprehensive ablation study to isolate the incremental contributions of distinct information sources. To establish a rigorous baseline, we pre-train the visual backbone using the DeepBDC framework (Xie et al., 2022). During inference, features are extracted directly via this frozen backbone and subsequently evaluated using a standard Logistic Regression (LR) classifier. Consequently, this baseline model relies exclusively on global image features, serving as a fundamental reference point prior to any multi-source augmentation. As illustrated in the table, we introduce two auxiliary components: base-class priors, which are retrieved from the base dataset to supplement scarce data, and local discriminative details, derived from multi-crop inputs to capture subtle visual cues. On the miniImageNet benchmark, the baseline attains an accuracy of 67.47% for the 5-way 1-shot task. The integration of individual auxiliary sources yields substantial improvements. Specifically, incorporating base-class priors boosts accuracy to 86.43%, demonstrating the efficacy of transferring prior knowledge to mitigate data scarcity. Similarly, integrating local details elevates performance to 85.64%, validating the significance of fine-grained patterns in distinguishing semantically similar classes. Most notably, the multi-source configuration, which fuses global, local, and base-class information, achieves superior performance across all datasets. For instance, in the miniImageNet 1-shot task, the multi-source model reaches 87.08%, representing a remarkable gain of 19.61% over the baseline. Even on the fine-grained CUB dataset, where local details are paramount, the multi-source fusion outperforms the strong local-only counterpart. These empirical findings collectively substantiate that global context, local details, and base-class priors offer complementary advantages that single-source paradigms cannot fully exploit. Crucially, the superiority of the

Table 3
Ablation study on the contributions of different information sources across the miniImageNet, tieredImageNet, and CUB datasets.

Dataset	Settings	Baseline	Base-source	Local-source	Multi-source
miniImageNet	5-way 1-shot	67.47 ± 0.84	86.43 ± 0.55	85.64 ± 0.61	**87.08 ± 0.58**
	5-way 5-shot	85.22 ± 0.52	91.12 ± 0.39	90.85 ± 0.45	**91.36 ± 0.39**
tieredImageNet	5-way 1-shot	72.64 ± 0.84	75.52 ± 0.91	75.38 ± 0.99	**78.50 ± 0.90**
	5-way 5-shot	88.89 ± 0.54	89.55 ± 0.55	89.38 ± 0.61	**90.04 ± 0.56**
CUB	5-way 1-shot	83.27 ± 0.75	86.25 ± 0.85	86.58 ± 0.79	**86.91 ± 0.78**
	5-way 5-shot	94.18 ± 0.32	94.72 ± 0.31	94.82 ± 0.31	**95.58 ± 0.28**

Table 4
The impact of different space constraints on the miniImageNet and tieredImageNet datasets.

Visual Proto Constraint	Semantic Proto Constraint	miniImageNet		tieredImageNet	
		5-way 1-shot	5-way 5-shot	5-way 1-shot	5-way 5-shot
✓		68.30 ± 0.45	88.63 ± 0.28	74.14 ± 0.83	89.09 ± 0.56
	✓	82.24 ± 0.42	79.74 ± 0.49	75.17 ± 0.98	82.03 ± 0.85
✓	✓	**87.08 ± 0.58**	**91.36 ± 0.39**	**78.50 ± 0.90**	**90.04 ± 0.56**

Table 5
The impact of external semantic priors on the miniImageNet and tieredImageNet datasets.

Dataset	Shot	DeepBDC + LR	Visual-MSIFN	Full-MSIFN	Δ Visual–DeepBDC	Δ Full–Visual
miniImageNet	1	67.471 ± 0.840	69.047 ± 0.439	87.080 ± 0.580	+1.576	+18.033
miniImageNet	5	85.223 ± 0.521	86.937 ± 0.269	91.360 ± 0.390	+1.714	+4.423
tieredImageNet	1	72.643 ± 0.841	73.921 ± 0.473	78.500 ± 0.900	+1.278	+4.579
tieredImageNet	5	88.895 ± 0.542	89.073 ± 0.302	90.040 ± 0.560	+0.178	+0.967

multi-source setting verifies the effectiveness of our Semantic-Guided Selection Mechanism. By rigorously screening for relevance, our framework effectively filters out noise, thereby preventing the negative transfer often associated with indiscriminate fusion. Instead, it successfully facilitates positive interaction among these heterogeneous sources to construct a robust and expressive representation. Ultimately, these results confirm that the rational utilization and strategic fusion of multi-source information constitute an effective pathway for enhancing few-shot learning performance.

4.4.2. Ablation of external semantic priors

To evaluate the contribution of external semantic priors, we compare three variants under the 5-way 1-shot and 5-way 5-shot settings on miniImageNet and tieredImageNet, as summarized in Table 5. DeepBDC + LR serves as a visual-only baseline, in which a logistic regression classifier is fitted to frozen global BDC features. Visual-MSIFN removes the external class-level semantic embeddings and all components that depend on them, while retaining the labeled base-class feature bank, local patch features, the Transformer-based fusion module, and the Visual Proto Constraint. For base-class matching and local-patch selection, this variant replaces semantic guidance with cosine similarity to visual class prototypes. Full-MSIFN further incorporates 300-dimensional GloVe class embeddings, a task-adapted cross-modal autoencoder, semantic-guided base-class matching and local-patch selection, and the Semantic Proto Constraint.

Visual-MSIFN achieves higher mean classification accuracy than DeepBDC + LR in all four settings, with gains ranging from 0.178 to 1.714 percentage points. This consistent pattern suggests that the visual components of MSIFN can exploit complementary base-class and local cues even without external semantic priors. Compared with Visual-MSIFN, Full-MSIFN yields additional accuracy gains ranging from 0.967 to 18.033 percentage points across the four settings. Notably, the gains are consistently larger in the 1-shot regime, suggesting that class-level semantic priors are especially useful when labeled support examples are scarce and visually derived class prototypes are less reliable.

4.4.3. Analysis of constraints

To evaluate the efficacy of the proposed Dual-Space Constraints in mitigating feature drift, we conduct an ablation study on both the miniImageNet and tieredImageNet datasets, with the results summarized in Table 4. This mechanism is designed to anchor fused representations to stable centroids within both the visual and semantic spaces.

When applied individually, the complementary nature of the two constraints becomes evident across different data regimes. Under the extreme data scarcity of the 1-shot setting, the Semantic Proto Constraint demonstrates exceptional efficacy. On the miniImageNet dataset, relying solely on this semantic guidance yields 82.24%, substantially outperforming the 68.30% achieved by the Visual Proto Constraint alone. A consistent, albeit narrower, advantage is observed on tieredImageNet (75.17% vs. 74.14%). This confirms that when visual evidence is severely limited and inherently noisy, the stability of high-level semantic embeddings provides reliable class-level guidance, effectively preventing the model from overfitting to the scarce visual samples.

Conversely, this dynamic completely reverses in the 5-shot scenario, where the Visual Proto Constraint significantly surpasses the Semantic Proto Constraint. This performance shift is attributed to the prototype computation process: averaging features from multiple support samples effectively suppresses instance-specific noise, yielding a highly precise visual centroid that accurately encapsulates the target class distribution. In contrast, static semantic embeddings, which represent high-level class abstractions, are less capable of capturing these rich, task-specific visual variations when sufficient visual samples become available.

Optimal performance is consistently achieved only when both constraints are enforced concurrently. The synergistic Dual-Space configuration establishes peak accuracies across all evaluated settings. Specifically, on miniImageNet, it elevates 1-shot and 5-shot performance to 87.08% and 91.36%, respectively. Correspondingly, tieredImageNet accuracies reach 78.50% and 90.04%. These empirical findings substantiate the highly complementary nature of the two constraints. By leveraging the visual constraint to capture task-specific visual details and the semantic constraint to preserve overall class consistency, the Dual-Space

mechanism ensures that the augmented representations remain robust and highly discriminative across varying data regimes.

4.4.4. Hyperparameter sensitivity analysis

To evaluate the sensitivity of MSIFN to the amount of auxiliary evidence retained during selection, we independently vary B and L on tieredImageNet in the 5-way 1-shot setting. Here, B denotes the number of base-class instances retrieved for each support example, and L denotes the number of local patches retained by the Semantic-Guided Selection Mechanism. The results are summarized in Fig. 3.

As shown in Fig. 3(a), accuracy rises from 77.83% at $B = 1$ to the best observed value of 78.54% at $B = 10$. Larger values of B yield no further improvement, and accuracy drops to 77.26% at $B = 30$. This trend suggests that a moderate number of visually similar base-class instances can provide complementary class-relevant evidence, whereas expanding the retrieval set admits lower-ranked and potentially mismatched neighbors that may dilute the contribution of relevant auxiliary information.

Fig. 3(b) shows a similar trade-off for L, with the best observed accuracy of 78.29% obtained at $L = 4$. When $L > 4$, accuracy remains lower, ranging from 76.91% to 77.82%. Because candidate patches are ranked by semantic relevance, retaining additional lower-ranked regions may introduce background clutter or class-irrelevant content. The results therefore favor retaining a compact set of semantically aligned patches rather than a larger set of less informative candidates.

Across the evaluated ranges, accuracy varies from 77.26% to 78.54% for B and from 76.91% to 78.29% for L, corresponding to variations of 1.28 and 1.38 percentage points, respectively. These variations suggest that MSIFN is only modestly sensitive to either hyperparameter over the tested ranges, although retaining excessive auxiliary information can reduce accuracy. Within these one-dimensional sweeps, the best observed settings are $B = 10$ and $L = 4$.

Table 6
Trainable parameter counts and peak allocated GPU memory for complete model variants; memory is measured on an NVIDIA A40.

Dataset / shot	Variant	Trainable parameters (M)	Peak GPU memory (GiB)
miniImageNet/1	DeepBDC + LR	12.548	0.924
miniImageNet/1	Visual-MSIFN	835.695	14.871
miniImageNet/1	Full-MSIFN	845.701	14.928
miniImageNet/5	Full-MSIFN	845.701	15.282
tieredImageNet/1	Full-MSIFN	845.701	22.773
tieredImageNet/5	Full-MSIFN	845.701	22.969

4.5. Computational cost and retrieval scalability

We evaluate MSIFN along three practical dimensions: complete-model computational and memory cost, the fidelity of approximate nearest-neighbor retrieval, and retrieval scalability under controlled hardware conditions.

Computational Complexity and Memory Footprint. We distinguish parameter counts for complete model variants from forward-pass complexity for individual components. Peak allocated GPU memory is measured during the execution of each complete variant on an NVIDIA A40, whereas operation counts use the convention of two FLOPs per MAC. The ResNet-12 backbone contains 12.465M trainable parameters and requires 3.523G MACs (7.046G FLOPs) per image. Including BDC pooling and projection increases the parameter count of DeepBDC + LR to 12.548M. Full-MSIFN contains 845.701M trainable parameters, of which 818.062M belong to the fusion Transformer.

Table 6 reports peak GPU memory for complete model variants. Because each value covers the execution of an entire variant, the alloca-

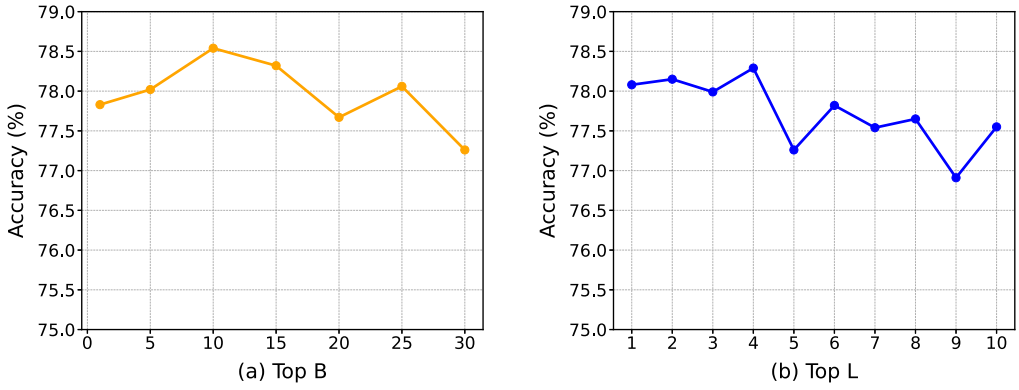


Fig. 3. Sensitivity of MSIFN to B and L on tieredImageNet under the 5-way 1-shot setting. (a) Classification accuracy as the number of retrieved base-class instances B varies. (b) Classification accuracy as the number of selected local patches L varies.

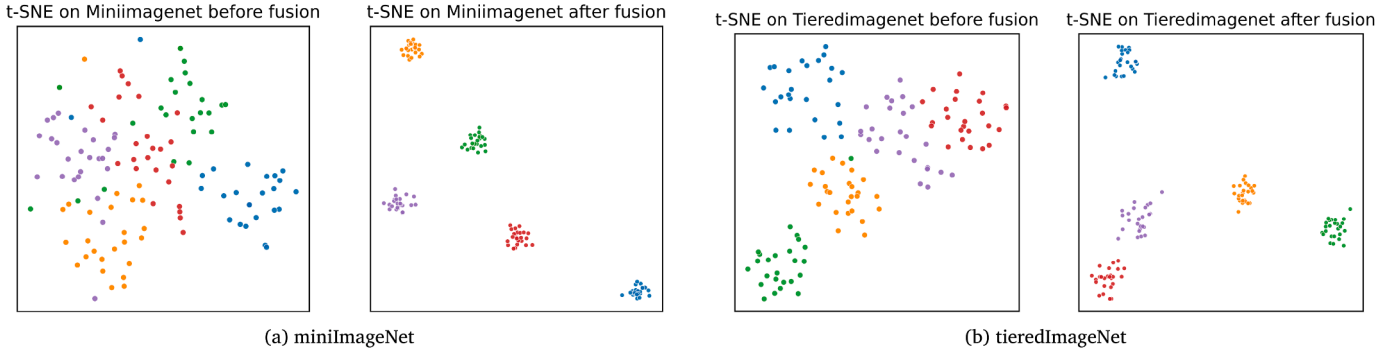


Fig. 4. The t-SNE visualization in the 5-way 5-shot setting on the miniImageNet and tieredImageNet datasets. Each circle represents a support sample, and different colors represent different classes.

Table 7
Paired classification accuracy and neighbor-retrieval agreement for GPU PyTorch Exact and CPU FAISS HNSW on fixed 2,000-episode manifests. These backend-specific results are not directly comparable with Table 1.

Dataset / shot	Exact accuracy (%)	HNSW accuracy (%)	HNSW − Exact (pp)	Recall@10	HNSW build time (s)	Serialized HNSW index size (GB)
miniImageNet/1	87.599	87.614	+0.015	98.268%	10.886	1.279
miniImageNet/5	91.225	91.273	+0.048	98.274%	12.365	1.279
tieredImageNet/1	78.939	78.988	+0.049	96.377%	247.654	14.940
tieredImageNet/5	90.310	90.343	+0.033	96.425%	209.013	14.940

tions cannot be decomposed additively by module. For Full-MSIFN, peak allocated memory ranges from 14.928 GiB for miniImageNet in the 1-shot setting to 22.969 GiB for tieredImageNet in the 5-shot setting. Together with the parameter and operation counts, these measurements indicate that the current implementation requires substantial computational and memory resources.

Approximate Retrieval Fidelity. By default, MSIFN uses exact nearest-neighbor retrieval to select the top-B instances within the matched base class according to squared Euclidean distance; all classification experiments use $B = 10$. HNSW search (Malkov & Yashunin, 2020) is considered only as an approximate deployment alternative. The CPU evaluations use FAISS (Johnson et al., 2019) Flat as the exact reference and FAISS HNSW as the approximate backend. As an implementation check, PyTorch Exact and FAISS Flat returned identical neighbor IDs for all ten evaluated queries. Recall@10 denotes the percentage of exact top-10 neighbor IDs recovered by HNSW and is distinct from recall in the classification task.

We select `efSearch` from $\{16, 32, 64, 128\}$ using 128 held-out base-feature queries per dataset, without test episodes or labels. The smallest value achieving at least 95% Recall@10 on all three datasets is `efSearch` = 16, which yields 98.906%, 97.344%, and 100.000% on miniImageNet, tieredImageNet, and CUB, respectively; CUB is used only in this selection step. Increasing `efSearch` to 128 raises Recall@10 to 100.000% on miniImageNet and tieredImageNet, while mean query latency increases from 0.610 to 1.039 ms and from 1.108 to 2.238 ms, respectively. These measurements characterize the retrieval recall–latency trade-off rather than classification performance. Unless stated otherwise, subsequent experiments use $M = 32$, `efConstruction` = 200, `efSearch` = 16, and four CPU threads.

Using this configuration, we compare GPU PyTorch Exact with CPU FAISS HNSW on identical fixed manifests of 2000 episodes, thereby controlling episode-sampling variation in the paired accuracy comparison. This backend study is separate from the primary evaluation in Table 1, so its absolute accuracies are interpretable only within the paired comparison. The backends were also run on different cluster nodes, precluding valid comparisons of retrieval latency and end-to-end wall-clock time.

As shown in Table 7, HNSW recovers 96.377%–98.274% of the exact top-10 neighbors, while the corresponding accuracy differences range from +0.015 to +0.049 percentage points. Because we report neither confidence intervals nor hypothesis tests for these paired differences, the results are descriptive and do not show that HNSW improves classification. Single-run HNSW construction times range from 10.886 to 247.654 s, and the serialized indexes occupy 1.279 GB for miniImageNet and 14.940 GB for tieredImageNet. Thus, the selected configuration provides high neighbor-set agreement but entails explicit construction and storage costs.

Same-Node Retrieval Scalability. We further conduct a same-node retrieval-only stress test of CPU FAISS Flat and HNSW. Both backends operate on identical pooled cross-class indexes containing 1k, 10k, or 50k candidates represented by 8,256-dimensional base features. All runs use four CPU threads and top-10 retrieval, with $M = 32$, `efConstruction` = 200, and `efSearch` = 16 for HNSW. For each feature source, 32 fixed queries are generated by perturbing source vectors with Gaussian noise of standard deviation 10^{-5}, and mean latency is computed over five timed passes. Random seeds are 1 for miniImageNet and 1001 for tieredImageNet. Because miniImageNet provides 38.4k source vectors, its 50k pool includes 11.6k perturbed copies. Each index is constructed once, and index size denotes the serialized footprint in decimal GB rather than peak memory or RSS. Table 8 summarizes the results.

As the candidate pool increases from 1k to 50k, FAISS Flat latency rises from 2.680 to 213.924 ms for miniImageNet-derived features and from 2.828 to 210.558 ms for tieredImageNet-derived features. HNSW latency increases more slowly, from 0.498 to 1.376 ms and from 0.663 to 1.682 ms, respectively, and remains lower at every evaluated scale. At 50k candidates, the serialized Flat and HNSW indexes occupy 1.6512 and 1.6648 GB, so HNSW adds approximately 13.6 MB beyond feature-vector storage. Single-run HNSW construction takes 105.648 and 161.844 s, compared with 1.002 and 0.395 s for Flat. For the tieredImageNet-derived 50k index, HNSW achieves 93.125% Recall@10, below the 95% selection target, showing that a tuned configuration may not retain its retrieval fidelity as index scale and candidate distribution change.

These findings apply only to the constructed workload, which does not reproduce the per-class candidate distribution used by MSIFN. Mean latency is estimated from 32 fixed queries, and each construction time is based on a single run. The experiment therefore supports lower same-node CPU query latency for HNSW under the evaluated conditions, but it does not establish the effect of approximate retrieval on classification accuracy or an end-to-end speedup for MSIFN.

4.6. Visualizations

To qualitatively assess the efficacy of MSIFN during task-specific adaptation, we visualize feature distributions using t-SNE on the miniImageNet and tieredImageNet datasets. Fig. 4 presents a comparative analysis between the baseline model and our MSIFN framework.

As illustrated in the baseline visualizations, the extracted features exhibit loosely clustered distributions with substantial inter-class overlap. Consequently, decision boundaries appear ambiguous, and intra-class variance remains high. This lack of clear separation highlights the insufficiency of relying solely on single-source global features, which struggle to maintain robust discriminability in data-scarce scenarios.

In contrast, the integration of multi-source information yields a marked improvement in the feature landscape. The learned embeddings form highly compact and well-separated clusters, demonstrating that MSIFN significantly enhances both intra-class compactness and inter-class separability. These visualizations corroborate our core premise: the strategic fusion of heterogeneous sources effectively suppresses noise and distills task-relevant cues, thereby constructing a robust feature space that facilitates accurate classification.

4.7. Limitations

Despite its promising performance, MSIFN depends on the reliability of class-level semantic priors. For miniImageNet and tieredImageNet,

Table 8
Same-node CPU retrieval scalability of FAISS Flat and HNSW on pooled cross-class indexes of 8,256-dimensional features. Mean query latency is computed over five timed passes with 32 fixed queries; this synthetic workload differs from the per-class retrieval used by MSIFN.

Feature source	Candidate count	Mean query latency (ms/query) Flat / HNSW	Index construction time (s) Flat / HNSW	Serialized index size (GB) Flat / HNSW
miniImageNet	1k	2.680/0.498	0.012/0.437	0.0330/0.0333
miniImageNet	10k	41.678/1.041	0.081/15.902	0.3302/0.3330
miniImageNet	50k	213.924/1.376	1.002/105.648	1.6512/1.6648
tieredImageNet	1k	2.828/0.663	0.008/0.431	0.0330/0.0333
tieredImageNet	10k	42.515/1.388	0.082/21.041	0.3302/0.3330
tieredImageNet	50k	210.558/1.682	0.395/161.844	1.6512/1.6648

these priors are GloVe embeddings; for CUB, they are human-annotated attributes. The framework assumes that semantic similarity serves as a useful proxy for visual similarity, an assumption that may break down when class names are ambiguous, semantic representations do not capture fine-grained visual distinctions, or a substantial domain shift separates base and novel classes. In these cases, bipartite matching may select a semantically related but visually mismatched base class, semantically guided local-patch ranking may retain class-irrelevant regions, and the Semantic Proto Constraint may anchor fused representations to a biased semantic target. Because we do not perturb the semantic priors in a controlled manner or evaluate cross-domain few-shot transfer, robustness to semantic noise and semantic–visual misalignment remains to be established.

MSIFN is also sensitive to support-set quality, particularly in the 1-shot setting. A support image affected by occlusion, degradation, or dominant background clutter may yield a biased global representation and few informative local patches. The Semantic-Guided Selection Mechanism can rank available candidates by relevance, but it cannot recover discriminative visual evidence missing from the support image. Moreover, the same support features drive the retrieval of auxiliary base-class instances and define the visual prototype used by the Visual Proto Constraint. A poor-quality support example can therefore compromise both feature augmentation and regularization. The current framework neither estimates the reliability of individual support examples nor accounts for uncertainty, and its robustness to corrupted support images has not been evaluated separately.

Finally, the current implementation has substantial computational and memory requirements. Full-MSIFN contains 845.701M trainable parameters and reaches a peak GPU memory allocation of 22.969 GiB (Table 6). It also performs episode-specific adaptation of the fusion Transformer and stores a persistent base-class feature bank for default exact retrieval. When HNSW is evaluated as an approximate deployment alternative, the serialized tieredImageNet index occupies 14.940 GB (Table 7), and single-run HNSW build times for the 50k-candidate stress tests range from 105.648 to 161.844 s (Table 8). These requirements may restrict deployment in latency-sensitive or resource-constrained environments. Although HNSW achieves lower mean CPU query latency than FAISS Flat in the same-node retrieval-only benchmark, this comparison uses pooled cross-class indexes and synthetically perturbed queries rather than MSIFN's per-class retrieval workload. Furthermore, the Exact and HNSW episode-level experiments were conducted on different cluster nodes. The reported results therefore characterize retrieval behavior and index overhead but do not establish a controlled end-to-end speedup for MSIFN.

Future work should investigate context-aware or domain-adapted semantic priors and assess their robustness under controlled semantic perturbations and cross-domain shifts. Other priorities include quality- and uncertainty-aware weighting of support examples and more efficient deployment through lower-dimensional representations, parameter-efficient task adaptation, and index compression. End-to-end studies should compare retrieval backends under identical hardware and software conditions and report the costs of index construction and updates, retrieval, task adaptation, and classification separately.

5. Conclusion

In this work, we propose the Multi-Source Information Fusion Network (MSIFN), a unified framework designed to alleviate the intrinsic feature bias stemming from data scarcity in few-shot learning. To overcome the limitations of relying on single modalities and the risks associated with indiscriminate fusion, our work advocates a paradigm of principled selection and strategic synthesis.

Specifically, we introduce a Semantic-Guided Selection Mechanism underpinned by a cross-modal autoencoder. This module facilitates intricate interactions among global contexts, local details, base-class priors, and semantic embeddings, enabling the model to adaptively distill high-quality, task-relevant cues while effectively suppressing background clutter. Furthermore, to counteract the feature drift associated with multi-source adaptation, we impose the Visual Proto Constraint and Semantic Proto Constraint. By anchoring fused representations to stable centroids within both the visual and semantic spaces, this dual-space mechanism ensures robust alignment. Extensive experiments on the miniImageNet, tieredImageNet, and CUB benchmarks demonstrate that MSIFN achieves superior performance over state-of-the-art methods.

Crucially, this study highlights that the principled fusion of heterogeneous sources is not merely a supplementary technique, but a fundamental paradigm shift for advancing few-shot learning. Looking ahead, the trajectory of multi-source fusion presents several compelling avenues. A primary direction involves transcending the limitations of static word embeddings by integrating richer, context-aware semantic priors from Large Language Models or vision-language foundation models. Concurrently, developing dynamic, modality-agnostic fusion architectures capable of adaptively weighting and integrating an arbitrary number of external knowledge sources represents a critical next step toward building more generalizable, robust, and open-world vision systems.

CRediT authorship contribution statement

Hantao Wang: Writing – original draft, Methodology, Formal analysis, Conceptualization; **Huan Wan:** Resources, Methodology; **Ghulan Mohiudin:** Investigation, Formal analysis; **Jianfeng Xu:** Supervision, Project administration; **Hong Rao:** Visualization, Validation; **Xin Wei:** Writing – review & editing, Validation, Supervision, Project administration, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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References

Afrasiyabi, A., Lalonde, J.-F., & Gagné, C. (2020). Associative alignment for few-shot image classification. In *European conference on computer vision* (pp. 18–35). Springer.

Cao, C., & Zhang, Y. (2022). Learning to compare relation: Semantic alignment for few-shot learning. *IEEE Transactions on Image Processing*, *31*, 1462–1474.

Chen, D., Zhang, J., Zheng, W.-S., & Wang, R. (2024). FeatWalk: Enhancing few-shot classification through local view leveraging. In *Proceedings of the AAAI conference on artificial intelligence* (pp. 1019–1027). (*vol. 38*).

Chen, W.-Y., Liu, Y.-C., Kira, Z., Wang, Y.-C. F., & Huang, J.-B. (2019). A closer look at few-shot classification. arxiv:1904.04232

Chen, Y., Liu, Z., Xu, H., Darrell, T., & Wang, X. (2021). Meta-baseline: Exploring simple meta-learning for few-shot learning. In *Proceedings of the IEEE/CVF international conference on computer vision* (pp. 9062–9071).

Devlin, J., Chang, M.-W., Lee, K., & Toutanova, K. (2019). BERT: Pre-training of deep bidirectional transformers for language understanding. In *Proceedings of the 2019 conference of the north american chapter of the association for computational linguistics: human language technologies, volume 1 (long and short papers)* (pp. 4171–4186). Association for Computational Linguistics.

Dong, M., Li, F., Li, Z., & Liu, X. (2025a). Contrastive prototype learning with semantic patchmix for few-shot image classification. *Engineering Applications of Artificial Intelligence*, *142*, 109917.

Dong, M., Li, F., Li, Z., & Liu, X. (2025b). PRSN: Prototype resynthesis network with cross-image semantic alignment for few-shot image classification. *Pattern Recognition*, *159*, 111122.

Dosovitskiy, A. (2020). An image is worth 16x16 words: Transformers for image recognition at scale. arxiv:2010.11929

Fei, N., Lu, Z., Xiang, T., & Huang, S. (2021). Melr: Meta-learning via modeling episode-level relationships for few-shot learning. In *International conference on learning representations*.

Finn, C., Abbeel, P., & Levine, S. (2017). Model-agnostic meta-learning for fast adaptation of deep networks. In *International conference on machine learning* (pp. 1126–1135). PMLR.

Fu, M., Wang, X., Wang, J., & Yi, Z. (2024). Prototype Bayesian meta-learning for few-shot image classification. *IEEE Transactions on Neural Networks and Learning Systems*, *36*(4), 7010–7024.

Johnson, J., Douze, M., & Jégou, H. (2019). Billion-scale similarity search with GPUs. *IEEE Transactions on Big Data*, *7*(3), 535–547.

Lai, J., Yang, S., Zhou, J., Wu, W., Chen, X., Liu, J., Gao, B.-B., & Wang, C. (2023). Clustered-patch element connection for few-shot learning. arxiv:2304.10093

Lee, K., Maji, S., Ravichandran, A., & Soatto, S. (2019). Meta-learning with differentiable convex optimization. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition* (pp. 10657–10665).

Li, J., Ye, C., Wang, F., & Pan, J. (2025a). A robust transductive distribution calibration method for few-shot learning. *Pattern Recognition*, *163*, 111488.

Li, W., Wang, L., Xu, J., Huo, J., Gao, Y., & Luo, J. (2019). Revisiting local descriptor based image-to-class measure for few-shot learning. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition* (pp. 7260–7268).

Li, X., Xue, S., Xie, J., Yang, X., Ma, Z., & Xue, J.-H. (2025b). Interactive triplet attention for few-shot fine-grained image classification. *Neurocomputing*, (p. 131377).

Li, Y., Guo, J., Qi, L., Li, W., & Shi, Y. (2025c). Text and image are mutually beneficial: Enhancing training-free few-shot classification with CLIP. In *Proceedings of the AAAI conference on artificial intelligence* (pp. 5039–5047). (*vol. 39*).

Liu, B., Cao, Y., Lin, Y., Li, Q., Zhang, Z., Long, M., & Hu, H. (2020). Negative margin matters: Understanding margin in few-shot classification. In *European conference on computer vision* (pp. 438–455). Springer.

Liu, F., Yang, S., Chen, D., Huang, H., & Zhou, J. (2024). Few-shot classification guided by generalization error bound. *Pattern Recognition*, *145*, 109904.

Ma, R., Fang, P., Drummond, T., & Harandi, M. (2022). Adaptive poincaré point to set distance for few-shot classification. In *Proceedings of the AAAI conference on artificial intelligence* (pp. 1926–1934). (*vol. 36*).

Malkov, Y. A., & Yashunin, D. A. (2020). Efficient and robust approximate nearest neighbor search using hierarchical navigable small world graphs. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, *42*(4), 824–836.

Padmanabhan, D. C., Gowda, S., Arani, E., & Zonooz, B. (2023). LSFSL: Leveraging shape information in few-shot learning. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition* (pp. 4971–4980).

Pan, M.-H., & Shen, H.-B. (2024). Cross-modal de-deviation for enhancing few-shot classification. *Pattern Recognition*, *152*, 110475.

Pan, M.-H., Xin, H.-Y., & Shen, H.-B. (2024). Semantic-based implicit feature transform for few-shot classification. *International Journal of Computer Vision*, *132*(11), 5014–5029.

Pennington, J., Socher, R., & Manning, C. D. (2014). Glove: Global vectors for word representation. In *Proceedings of the 2014 conference on empirical methods in natural language processing (EMNLP)* (pp. 1532–1543).

Radford, A., Kim, J. W., Hallacy, C., Ramesh, A., Goh, G., Agarwal, S., Sastry, G., Askell, A., Mishkin, P., Clark, J., Krueger, G., & Sutskever, I. (2021). Learning transferable visual models from natural language supervision. In *Proceedings of the 38th international conference on machine learning* (pp. 8748–8763). PMLR (*vol. 139*). Proceedings of Machine Learning Research.

Ravi, S., & Larochelle, H. (2017). Optimization as a model for few-shot learning. In *International conference on learning representations*.

Ren, L., Duan, G., Huang, T., & Kang, Z. (2022). Multi-local feature relation network for few-shot learning. *Neural Computing and Applications*, *34*(10), 7393–7403.

Ren, M., Triantafillou, E., Ravi, S., Snell, J., Swersky, K., Tenenbaum, J. B., Larochelle, H., & Zemel, R. S. (2018). Meta-learning for semi-supervised few-shot classification. In *International conference on learning representations*.

Russakovsky, O., Deng, J., Su, H., Krause, J., Satheesh, S., Ma, S., Huang, Z., Karpathy, A., Khosla, A., Bernstein, M. et al. (2015). Imagenet large scale visual recognition challenge. *International journal of computer vision*, *115*(3), 211–252.

Schwartz, E., Karlinsky, L., Feris, R., Giryes, R., & Bronstein, A. (2022). Baby steps towards few-shot learning with multiple semantics. *Pattern Recognition Letters*, *160*, 142–147.

Snell, J., Swersky, K., & Zemel, R. (2017). Prototypical networks for few-shot learning. *Advances in neural information processing systems*, *30*, 4077–4087.

Sun, S., & Gao, H. (2023). Meta-Adam: An meta-learned adaptive optimizer with momentum for few-shot learning. *Advances in Neural Information Processing Systems*, *36*, 65441–65455.

Sun, Z., Zheng, W., Guo, P., & Wang, M. (2025). Tst_mfl: Two-stage training based metric fusion learning for few-shot image classification. *Information Fusion*, *113*, 102611.

Tian, Y., Wang, Y., Krishnan, D., Tenenbaum, J. B., & Isola, P. (2020). Rethinking few-shot image classification: A good embedding is all you need? In *European conference on computer vision* (pp. 266–282). Springer.

Vinyals, O., Blundell, C., Lillicrap, T., Wierstra, D. et al. (2016). Matching networks for one shot learning. *Advances in Neural Information Processing Systems*, *29*, 3630–3638.

Wah, C., Branson, S., Welinder, P., Perona, P., & Belongie, S. (2011). The caltech-ucsd birds-200-2011 dataset. Computation & Neural Systems Technical Report CNS-TR-2011-001, California Institute of Technology.

Wang, F., Xiang, X., Cheng, J., & Yuille, A. L. (2017). NormFace: L2 hypersphere embedding for face verification. In *Proceedings of the 25th ACM international conference on multimedia* (pp. 1041–1049).

Wang, H., Pang, G., Wang, P., Zhang, L., Wei, W., & Zhang, Y. (2023a). Glocal energy-based learning for few-shot open-set recognition. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition* (pp. 7507–7516).

Wang, Q.-W., Zhou, D.-W., Zhang, Y.-K., Zhan, D.-C., & Ye, H.-J. (2023b). Few-shot class-incremental learning via training-free prototype calibration. *Advances in Neural Information Processing Systems*, *36*, 15060–15076.

Wang, X., Wang, X., Jiang, B., & Luo, B. (2023c). Few-shot learning meets transformer: Unified query-support transformers for few-shot classification. *IEEE Transactions on Circuits and Systems for Video Technology*, *33*(12), 7789–7802.

Wei, X., Du, W., Wan, H., & Min, W. (2023). Feature distribution fitting with direction-driven weighting for few-shot images classification. In *Proceedings of the AAAI conference on artificial intelligence* (pp. 10315–10323). (*vol. 37*).

Wertheimer, D., Tang, L., & Hariharan, B. (2021). Few-shot classification with feature map reconstruction networks. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition* (pp. 8012–8021).

Xiao, T., Xia, Y., Tang, R., Du, W., & Wang, Z. (2025). Fusion of global and adaptive local information for few-shot image classification. *Pattern Recognition*, *168*, 111802.

Xie, J., Long, F., Lv, J., Wang, Q., & Li, P. (2022). Joint distribution matters: Deep brownian distance covariance for few-shot classification. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition* (pp. 7972–7981).

Xu, J., Wang, B., Wan, H., Su, P., & Wei, X. (2025). Adaptive blank compensation for few-shot image classification. *Neurocomputing*, *637*, 130127.

Yang, F., Wang, R., & Chen, X. (2022). Sega: Semantic guided attention on visual prototype for few-shot learning. In *Proceedings of the IEEE/CVF winter conference on applications of computer vision* (pp. 1056–1066).

Yang, F., Wang, R., & Chen, X. (2023). Semantic guided latent parts embedding for few-shot learning. In *Proceedings of the IEEE/CVF winter conference on applications of computer vision* (pp. 5447–5457).

Yang, S., Wu, S., Liu, T., & Xu, M. (2021). Bridging the gap between few-shot and many-shot learning via distribution calibration. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, *44*(12), 9830–9843.

Zhang, B., Luo, C., Yu, D., Li, X., Lin, H., Ye, Y., & Zhang, B. (2024a). MetaDiff: Meta-learning with conditional diffusion for few-shot learning. In *Proceedings of the AAAI conference on artificial intelligence* (pp. 16687–16695). (*vol. 38*).

Zhang, C., Cai, Y., Lin, G., & Shen, C. (2020). DeepEMD: Few-shot image classification with differentiable earth mover's distance and structured classifiers. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition* (pp. 12203–12213).

Zhang, H., Xu, J., Jiang, S., & He, Z. (2024b). Simple semantic-aided few-shot learning. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition* (pp. 28588–28597).

Zhang, J., Liu, W., Zhao, Z., & Su, F. (2021). Distribution estimation based pseudo-feature library generation for few-shot image classification. In *2021 IEEE International conference on multimedia & expo workshops (ICMEW)* (pp. 1–6). IEEE.

Zhang, R., Tan, J., Cao, Z., Xu, L., Liu, Y., Si, L., & Sun, F. (2024c). Part-aware correlation networks for few-shot learning. *IEEE Transactions on Multimedia*, *26*, 9527–9538.

Zhang, T., & Huang, W. (2022). Kernel relative-prototype spectral filtering for few-shot learning. In *European conference on computer vision* (pp. 541–557). Springer.

Zhou, J., & Cai, Q. (2024). Global and local representation collaborative learning for few-shot learning. *Journal of Intelligent Manufacturing*, *35*(2), 647–664.

Zhu, Y., Xie, J., Tang, Z., Peng, X., & Elgammal, A. (2019). Semantic-guided multi-attention localization for zero-shot learning. *Advances in Neural Information Processing Systems*, *32*, 14917–14927.